Simulation Assumption Uncertainty Module (SAUM)

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Independent Methodological Authority

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Governed Space

Simulation Assumption Uncertainty

Canonical Definition

Simulation Assumption Uncertainty Module (SAUM) defines the governance framework for identifying, characterizing, bounding, classifying, propagating, reassessing, and preserving uncertainty associated with Simulation Assumptions throughout the simulation lifecycle.

It establishes the methodological conditions under which uncertainty attached to an assumption remains explicitly connected to the assumption, the simulation context in which that assumption operates, the dependencies through which its influence propagates, and the simulation outputs and conclusions that may be materially affected by it.

SAUM does not attempt to eliminate uncertainty.

It governs what is uncertain about a Simulation Assumption, why that uncertainty exists, how significant it may be within the simulation, where its influence may propagate, and how that uncertainty must remain visible rather than being converted into false certainty.

Operational Role

SAUM serves as the fourth internal governance space of the Simulation Assumption Standard (SAS).

The preceding modules establish:

SAIM — Which assumptions exist and which are materially relevant?

SABM — On what basis is each assumption being used?

SADM — What depends upon each assumption?

SAUM asks:

What uncertainty exists around each Simulation Assumption, how should that uncertainty be represented, and where can it materially influence the simulation?

The governed progression is:

Governed Simulation Assumption → Uncertainty Identification → Uncertainty Source → Uncertainty Characterization → Uncertainty Bound → Dependency Connection → Uncertainty Propagation → Materiality → Reassessment → Uncertainty Record

The resulting uncertainty structure becomes an input to:

SALM — Simulation Assumption Limitation Module

Module Operational Space

SAUM governs: assumption uncertainty identification, uncertainty source identification, uncertainty characterization, uncertainty classification, quantitative uncertainty, qualitative uncertainty, bounded uncertainty, unbounded uncertainty, known uncertainty, partially known uncertainty, unresolved uncertainty, epistemic uncertainty, variability-related uncertainty, uncertainty ranges, uncertainty distributions where applicable, uncertainty confidence, uncertainty sensitivity, uncertainty materiality, uncertainty dependency, uncertainty interaction, uncertainty accumulation, uncertainty amplification, uncertainty dampening, uncertainty propagation, compound uncertainty, correlated uncertainty, uncertainty concentration, uncertainty thresholds, uncertainty change, uncertainty reassessment, uncertainty disclosure, and complete assumption-uncertainty traceability.

Module Function

Simulation Assumptions are often necessary precisely because complete knowledge is unavailable.

An assumption may represent: an estimated future condition, an incompletely observed physical property, an uncertain human behavior, an environmental condition, an engineering approximation, an unknown operational response, an incomplete dataset condition, or another state that cannot be represented with complete certainty.

The existence of an assumption therefore does not imply that its uncertainty has been sufficiently governed.

SAUM transforms:

Assumption

into:

Assumption + Uncertainty + Uncertainty Basis + Bound + Dependency Context + Propagation Consequence

The purpose is to prevent uncertainty from disappearing merely because an uncertain condition has been expressed as a fixed simulation assumption.

Simulation Assumption Uncertainty

Simulation Assumption Uncertainty is the governed lack of complete certainty concerning the truth, value, range, stability, applicability, behavior, or future state represented by a Simulation Assumption.

The relevant uncertainty must be understood relative to:

Simulation Object + Purpose + Scope + Reality Representation + Assumption + Simulation Context

An assumption may therefore carry different uncertainty significance in different simulations.

Uncertainty Identification

Every material Simulation Assumption should be examined for uncertainty.

The inquiry asks:

What about this assumption is not known with sufficient certainty?

The answer may concern: the assumed value, expected range, probability, stability, timing, frequency, applicability, underlying evidence, environmental conditions, behavioral response, or another materially relevant property.

Uncertainty Source

SAUM distinguishes the assumption from the source of its uncertainty.

Uncertainty may arise because: measurements are incomplete, evidence is limited, future conditions are unknown, observed populations are incomplete, physical variability exists, behavior varies, environmental states change, models simplify reality, historical information is sparse, expert judgment differs, dependencies are incompletely understood, or relevant knowledge does not yet exist.

Material uncertainty should remain attributable to its identifiable source where possible.

Known Uncertainty

A Known Uncertainty exists where the existence and relevant nature of uncertainty are recognized.

Example:

Wind speed is expected to vary between defined operational bounds.

The exact future wind speed is unknown, but the uncertainty itself is understood sufficiently to be represented.

Partially Known Uncertainty

A Partially Known Uncertainty exists where some properties of the uncertainty are understood while others remain unresolved.

For example:

a likely range may be known while the probability distribution remains uncertain.

The unresolved portion must remain visible.

Unresolved Uncertainty

An Unresolved Uncertainty exists where material uncertainty is known to exist but cannot yet be sufficiently characterized or bounded.

SIMULOS® establishes:

Unknown magnitude does not justify zero uncertainty.

Where uncertainty cannot be sufficiently quantified, it should remain explicitly represented as unresolved.

Quantitative Uncertainty

Where justified, uncertainty may be represented quantitatively through: ranges, intervals, distributions, probabilities, tolerances, confidence bounds, sensitivity values, or another applicable quantitative representation.

The representation should not imply greater precision than the underlying uncertainty basis supports.

Qualitative Uncertainty

Not all uncertainty can legitimately be reduced to numbers.

Qualitative uncertainty may be represented through governed classifications such as:

Low

Moderate

High

Critical

or another defined classification scheme.

The classification basis should remain explicit.

No False Quantification Principle

SIMULOS® establishes:

Uncertainty must not be assigned numerical precision merely to make the simulation appear more rigorous.

Where the basis supports only qualitative characterization, quantitative representation should not manufacture unsupported certainty.

Bounded Uncertainty

A Bounded Uncertainty exists where an assumption can reasonably be constrained within defined limits.

Example:

Temperature Assumption: 18–25°C

The bound should identify: lower limit, upper limit, basis, applicable context, and conditions under which the bound remains meaningful.

Unbounded Uncertainty

Some uncertainty cannot be sufficiently bounded.

Where a material assumption contains unbounded uncertainty, SAUM requires that condition to remain visible.

Unbounded uncertainty may materially constrain: simulation interpretation, scenario confidence, output confidence, or the legitimate strength of conclusions.

SAUM identifies the condition without making the external validation determination.

Uncertainty Range

An Assumption Uncertainty Range defines the governed interval within which the relevant assumed condition is reasonably expected to vary.

A range should not automatically be interpreted as:

all values equally likely

unless that interpretation is independently justified.

Uncertainty Distribution

Where sufficient basis exists, uncertainty may be represented through a probability distribution.

The distribution itself may depend upon assumptions.

SAUM therefore requires the basis of the distribution to remain distinguishable from the uncertainty being represented.

Uncertainty Confidence

A simulation may possess uncertainty not only about an assumed value, but about the uncertainty estimate itself.

Example:

Assumed Range: 10–15

may itself have:

Low Confidence

if based upon limited observations.

SAUM permits this second-order uncertainty to remain visible rather than treating the range as absolute.

Epistemic Uncertainty

Epistemic Uncertainty arises from incomplete knowledge.

It may potentially be reduced through: better evidence, additional measurement, improved observation, further research, or stronger understanding.

Examples include uncertainty concerning: material properties, system behavior, causal relationships, or poorly observed conditions.

Variability-Related Uncertainty

Some uncertainty arises because reality itself varies.

Examples include: weather, human behavior, traffic, biological response, market behavior, manufacturing variability, or environmental fluctuation.

This uncertainty may persist even where extensive information exists.

SAUM preserves the distinction where it materially affects simulation governance.

Uncertainty Materiality

Not every uncertainty requires the same governance depth.

Materiality may depend upon: uncertainty magnitude, affected assumption, dependency breadth, dependency depth, affected model components, sensitivity, output consequence, scenario consequence, and intended simulation use.

The governing question is:

Could this uncertainty materially change simulation behavior, outputs, or conclusions?

Uncertainty Sensitivity

An assumption may contain substantial uncertainty while the simulation remains relatively insensitive to it.

Another assumption may contain only small uncertainty while simulation outputs are highly sensitive to small changes.

SAUM therefore separates:

Uncertainty Magnitude

from:

Simulation Sensitivity

Conceptually:

Small Uncertainty + High Sensitivity

may be more material than:

Large Uncertainty + Low Sensitivity

Uncertainty Dependency

SADM establishes what depends upon an assumption.

SAUM uses that dependency structure to determine where assumption uncertainty may matter.

Conceptually:

Assumption

↓

Assumption Uncertainty

↓

Dependency Path

↓

Affected Simulation Element

↓

Affected Output Uncertainty Propagation

Simulation Assumption Uncertainty Propagation occurs where uncertainty attached to an assumption influences downstream simulation elements.

Example:

Uncertain Road Friction

↓

Braking Model

↓

Stopping Distance

↓

Collision Probability

The uncertainty does not disappear merely because it passes through a model.

No Uncertainty Laundering Principle

SIMULOS® establishes:

Uncertainty attached to a material assumption does not become certainty merely because the assumption passes through mathematical, computational, or simulation processing.

Transformation may change the uncertainty.

It does not automatically eliminate it.

Uncertainty Amplification

A simulation dependency may amplify assumption uncertainty.

Example:

±1% uncertainty in Input Assumption

may produce:

±15% variation in Output

where the model is highly sensitive.

Such amplification should remain identifiable.

Uncertainty Dampening

Simulation structure may reduce the downstream effect of assumption uncertainty.

Where demonstrated, SAUM may record that uncertainty is dampened through the dependency path.

Dampening should be demonstrated rather than presumed.

Uncertainty Accumulation

Multiple assumptions may contribute uncertainty to the same simulation element or output.

Conceptually:

Assumption A Uncertainty Assumption B Uncertainty Assumption C Uncertainty

↓

Output Uncertainty

SAUM requires material accumulation to remain visible.

Correlated Uncertainty

Uncertainties may not be independent.

Two assumptions may derive from: the same evidence, the same environmental condition, the same dataset, the same expert judgment, or the same underlying physical process.

Treating correlated uncertainty as independent may materially misrepresent simulation confidence.

SAUM therefore requires material correlation to be identified where known.

Compound Uncertainty

A Compound Uncertainty exists where several uncertain assumptions interact in a way that cannot be adequately understood by evaluating each one independently.

This may occur in: nonlinear systems, coupled models, multi-agent simulations, complex environmental models, or highly interdependent systems.

Uncertainty Concentration

A simulation may contain many assumptions while most material uncertainty is concentrated in only a small number.

SAUM may identify:

High-Uncertainty Assumption Concentration

where a limited set of assumptions dominates uncertainty in key outputs.

This can become an important simulation-governance signal.

Uncertainty Interaction

Assumption uncertainties may interact.

For example:

uncertainty in:

Reaction Time

and:

Road Friction

may jointly influence collision outcomes more strongly than either uncertainty considered independently.

Material interactions should remain governable.

Uncertainty Threshold

A simulation may establish thresholds beyond which assumption uncertainty becomes materially incompatible with the intended simulation purpose.

For example:

Uncertainty ≤ Threshold X

may be acceptable for exploratory simulation.

The same uncertainty may be unacceptable for high-consequence simulation.

SAUM therefore treats thresholds as context-dependent governance conditions.

Uncertainty and Simulation Depth

Required uncertainty governance may increase with: simulation consequence, simulation complexity, decision significance, required precision, reliance expectations, and Simulation Depth.

A low-consequence exploratory simulation may legitimately tolerate broader uncertainty.

A simulation supporting high-consequence engineering decisions may require substantially stronger characterization.

Uncertainty and Precision

Simulation output precision should not be confused with uncertainty.

A simulation may output:

12.347821

while the governing assumption supports only:

approximately 10–15

Numerical output precision does not eliminate upstream uncertainty.

No Precision Inflation Principle

SIMULOS® establishes:

Simulation output precision must not be interpreted as greater certainty than the assumptions and governed simulation basis can support. Uncertainty vs. Assumption Basis

SABM governs:

Why is the assumption being used?

SAUM governs:

What remains uncertain about that assumption?

A strong assumption basis may still contain material uncertainty.

The two governance questions are distinct. Uncertainty vs. Dependency

SADM governs:

What depends upon the assumption?

SAUM governs:

How can uncertainty associated with that assumption influence those dependencies?

This preserves clean module boundaries. Uncertainty vs. Limitation

SAUM governs uncertainty.

SALM subsequently governs:

Where the assumption ceases to remain applicable, what constraints it creates, and which limitations must remain visible.

Uncertainty may create a limitation.

It is not itself identical to limitation governance. Uncertainty vs.
Validation

SAUM does not determine whether a Simulation Assumption is universally valid or whether a simulation may support an external validation claim.

Those questions belong to the appropriate validation governance under VALIDOS®.

SAUM supplies the governed uncertainty representation needed for such downstream evaluation where applicable. Uncertainty vs. Integrity

SAUM does not replace INTEGROS®.

INTEGROS® may govern whether relevant uncertainty records, evidence, states, or processes preserve required integrity.

SAUM governs specifically the uncertainty associated with Simulation Assumptions inside the simulation governance lifecycle.

Uncertainty Change

Assumption uncertainty may change when: new evidence appears, measurements improve, the simulation environment changes, the Simulation Object changes, the assumption is revised, dependency relationships change, new scenarios are introduced, or new uncertainty is discovered.

The uncertainty state should therefore remain capable of reassessment.

Uncertainty Reduction

New evidence or improved knowledge may reduce uncertainty.

SAUM permits:

Previous Uncertainty → New Basis → Reduced Uncertainty

The historical uncertainty should remain traceable to the simulation versions for which it applied.

Uncertainty Expansion

New evidence may reveal that prior uncertainty was underestimated.

SAUM permits:

Previous Bound → New Evidence → Expanded Bound

Expansion is not a governance failure.

Failure would occur if material new uncertainty were concealed while the previous narrower representation continued to be used.

Uncertainty Reclassification

An uncertainty may move between states such as:

Unresolved → Partially Characterized → Bounded

or:

Bounded → Unresolved

when new information changes the available basis.

Uncertainty Change Trigger

Reassessment may be triggered by: assumption modification, basis modification, dependency modification, model change, environment change, new evidence, contradictory evidence, new scenarios, unexpected simulation behavior, or changed simulation purpose and scope.

Uncertainty Version Binding

Every material uncertainty representation should remain bound to the relevant: Simulation Object, assumption version, simulation configuration, model version, environment version, and applicable lifecycle state

where these distinctions materially affect the uncertainty.

Historical Uncertainty Preservation

Later reduction of uncertainty should not erase the fact that earlier simulation outputs were produced under greater uncertainty.

Historical uncertainty records support: simulation reconstruction, comparison, audit, learning, and lifecycle accountability.

Uncertainty Gap

A Simulation Assumption Uncertainty Gap exists where material uncertainty: has not been identified, cannot be characterized, lacks sufficient basis, has unknown bounds, has unknown propagation, has unresolved correlation, or cannot be connected to affected simulation elements.

The gap itself should become a governed condition.

No Unknown-to-Zero Principle

SIMULOS® establishes:

Uncertainty that cannot currently be determined must not be represented as zero uncertainty.

The correct state may be:

Uncertainty Undetermined

rather than:

No Uncertainty Uncertainty Disclosure

Material assumption uncertainty should remain visible to downstream simulation governance.

Disclosure may include: uncertainty type, magnitude or classification, basis, confidence, unresolved elements, affected dependencies, affected outputs, and applicable conditions.

Uncertainty Traceability

SAUM preserves the chain:

Simulation Assumption

↓

Uncertainty Source

↓

Uncertainty Characterization

↓

Uncertainty Bound / Status

↓

Dependency Path

↓

Affected Simulation Element

↓

Affected Output / Conclusion

where applicable.

Simulation Assumption Uncertainty Record

For each material assumption uncertainty, SAUM may establish a record containing: Uncertainty ID, Simulation Assumption ID, uncertainty description, uncertainty source, uncertainty type, quantitative or qualitative representation, lower and upper bounds where applicable, distribution where applicable, confidence, materiality, sensitivity, dependency references, propagation path, correlation references, interaction references, threshold status, unresolved uncertainty, uncertainty change history, simulation version, and traceability references.

Assumption Uncertainty Map

Material uncertainties may be represented through a Simulation Assumption Uncertainty Map connecting:

Assumptions → Uncertainties → Dependencies → Affected Outputs

The map allows simulation operators and reviewers to identify where uncertainty is concentrated and where it may materially influence simulation outcomes.

Uncertainty Status

An assumption uncertainty may be classified as:

Identified

Characterized

Bounded

Partially Bounded

Unbounded

Conditional

Material

Critical

Under Reassessment

Resolved

or:

Undetermined

according to the applicable governance context.

Minimum Implementation Framework

1. Identify Material Assumption Uncertainty

Examine each material Simulation Assumption for uncertainty relevant to the simulation. 2. Identify the Uncertainty Source

Determine why the uncertainty exists and what information supports its characterization. 3. Characterize the Uncertainty

Determine whether the uncertainty is quantitative, qualitative, bounded, partially bounded, unbounded, known, or unresolved. 4. Establish Appropriate Bounds

Where justified, define ranges, distributions, tolerances, confidence levels, or other appropriate representations. 5. Connect Uncertainty to Dependencies

Use the governed dependency structure to identify which simulation elements may be affected. 6. Assess Materiality and Sensitivity

Determine whether the uncertainty can materially influence model behavior, scenarios, outputs, or conclusions. 7. Assess Interaction and Accumulation

Identify material correlation, compound uncertainty, concentration, amplification, or interaction between uncertain assumptions. 8. Trace Uncertainty Propagation

Preserve how material uncertainty can propagate through the simulation. 9. Record Uncertainty Gaps

Keep unresolved or insufficiently characterized material uncertainty visible. 10. Maintain Uncertainty Lifecycle Traceability

Bind uncertainty representations to the relevant assumption and simulation versions and reassess them when material change occurs.

Governance Outputs

SAUM may produce: Simulation Assumption Uncertainty Identifier, Simulation Assumption Uncertainty Record, Uncertainty Source Record, Uncertainty Characterization Record, Quantitative Uncertainty Record, Qualitative Uncertainty Record, Uncertainty Bound, Uncertainty Range Record, Uncertainty Distribution Record, Uncertainty Confidence Record, Unresolved Uncertainty Record, Uncertainty Materiality Assessment, Uncertainty Sensitivity Record, Uncertainty Dependency Record, Uncertainty Propagation Path, Uncertainty Amplification Record, Uncertainty Dampening Record, Uncertainty Accumulation Record, Correlated Uncertainty Record, Compound Uncertainty Record, Uncertainty Interaction Record, Uncertainty Concentration Record, Uncertainty Threshold Record, Uncertainty Change Record, Uncertainty Reassessment Record, Uncertainty Gap Record, Simulation Assumption Uncertainty Map, Simulation Assumption Uncertainty Traceability Chain, and Governed Simulation Assumption Uncertainty Set.

Use Case 1 — Autonomous Vehicle Simulation

Scenario

A vehicle simulation assumes:

Average human pedestrian reaction behavior remains within a defined response range.

The assumption is required for collision-avoidance scenarios. Application

SAUM identifies that the response range contains substantial human variability.

The simulation is highly sensitive to the upper tail of delayed pedestrian reactions.

The uncertainty is therefore connected through:

Pedestrian Response Assumption

↓

Behavioral Model

↓

Collision-Avoidance Scenario

↓

Stopping / Evasion Outcome Result

The assumption is classified as:

Material Uncertainty — High Output Sensitivity

The uncertainty remains visible in the simulation record rather than being hidden behind one fixed pedestrian-response value. Use Case 2 — Space Mission Simulation

Scenario

A mission simulation assumes a defined degradation rate for a spacecraft power subsystem over a multi-year mission. Application

The degradation rate contains uncertainty resulting from: limited long-duration observations, radiation exposure variability, temperature cycles, and manufacturing variability.

SAUM maps uncertainty propagation through:

Degradation Assumption

↓

Power Generation

↓

Battery Availability

↓

Thermal / Communication Capacity

↓

Mission Availability Result

The simulation records a bounded uncertainty range together with a residual unresolved component.

Mission outputs cannot therefore represent the degradation assumption as a perfectly known future state.

Architectural Position

Simulation Assumption Uncertainty is the fourth internal governance space of the Simulation Assumption Standard (SAS).

The complete SAS progression is:

Simulation Assumption Identification → Simulation Assumption Basis → Simulation Assumption Dependency → Simulation Assumption Uncertainty → Simulation Assumption Limitation

SAIM establishes:

Which assumptions exist?

SABM establishes:

Why are they being used?

SADM establishes:

What depends upon them?

SAUM establishes:

What remains uncertain about them and where can that uncertainty propagate?

SALM establishes:

Where do those assumptions cease to remain applicable and what limitations follow?

Together, the five modules ensure that Simulation Assumptions cannot function as invisible premises, unsupported simplifications, isolated model conditions, falsely certain inputs, or unbounded representations of reality.

Foundational Principle

Uncertainty does not disappear when an uncertain condition becomes a Simulation Assumption; it must remain identifiable, bounded where possible, connected to its dependencies, and traceable through every materially affected simulation outcome.

Canonical Closing Statement

Simulation Assumption Uncertainty Module establishes the uncertainty-governance layer of the Simulation Assumption Standard by identifying, characterizing, bounding, and preserving uncertainty associated with Simulation Assumptions and tracing its influence through the dependencies, models, scenarios, outputs, and conclusions that may materially depend upon those assumptions. Through uncertainty-source identification, quantitative and qualitative characterization, bounds, confidence, sensitivity, materiality, correlation, interaction, accumulation, amplification, propagation, change reassessment, gap visibility, version binding, and complete uncertainty traceability, SAUM prevents uncertain simulation premises from being transformed into false certainty merely through computational processing and ensures that simulation outcomes remain connected to the uncertainty embedded in the assumptions from which they were produced.

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